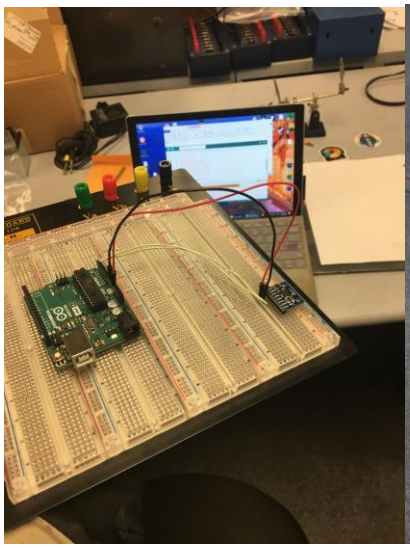
Earthquake Tower Construction

Introduction:

For the STLCC open house, the Engineering club had to build a tower that could withstand a scaled down earthquake (simulated using a reciprocating arm and rollers. The tower had to be 3ft tall and stand on a wooden base; the tower was also to use a tuned mass dampener.

Design:

A standard Arduino UNO was used to monitor a six-axis accelerometer gyroscope hybrid chip, used to gather graphable acceleration data. The tower was built in stages, from 1in x 1ft wooden dowels (for the primary beams) and 1/8thin x 1ft wooden dowels (for the cross beams). First, a 2ft tall prototype was constructed to test the feasibility of the whole system. After that the tower was dismantled and assembled using standard cross bracing and tensioners, and tested, the result was a solid structure that would completely contain earthquake shocks (and remain rigid even when held horizontal), so well in-fact the TMD was useless, and so all of the braces had to be removed. The removal of the braces afforded enough elasticity to let the dampener reduce the structure shaking.





Conclusion:

The exhibit was successful; it provided a visual and easy-to-grasp example of structural engineering and tuned mass dampeners.